Block practice · Breathing and gas exchange

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Name thorax and airway structures and the route of air

Explain how diaphragm and intercostal muscles drive breathing

Explain how smoking damages health, including in pregnancy

Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Explain how alveoli are adapted for gas exchange and effects of damage

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Practice workbook · Name thorax and airway structures and the route of air

Trachea, bronchi, bronchioles, alveoli, ribs, intercostal muscles, sternum, diaphragm on diagrams and models.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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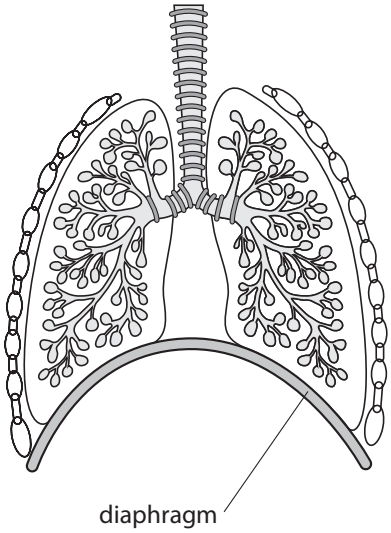
Type ID: HBI-THORAX-STRUCTURE-NAME

1. Question 4 · part a.i · 3 marks

June 2024 Paper 2

Worked-example retry · Attempt part a.i only.

4 (a) The diagram shows the human lungs.



(i) Add the following labels on the diagram

(3)

- trachea
- bronchus
- alveoli

(ii) Give four features of the lungs that adapt them for gas exchange.

(4)

1 .....

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2 .....

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3 .....

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4 .....

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1. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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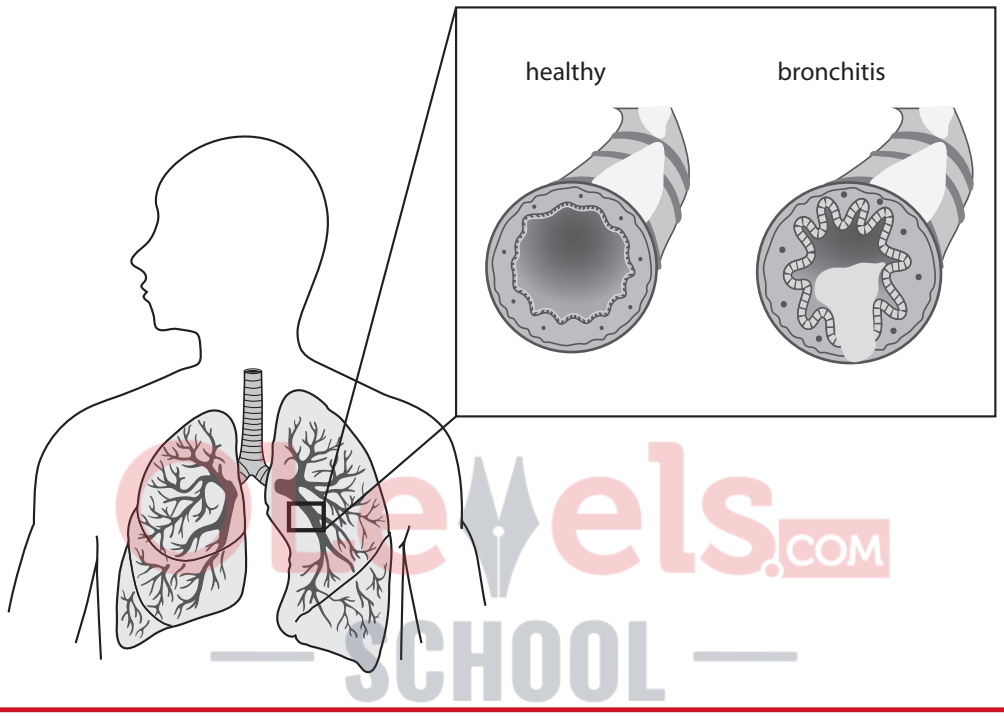
2. Question 6 · part a · 1 mark

January 2020 Paper 1

Further practice · Attempt part a only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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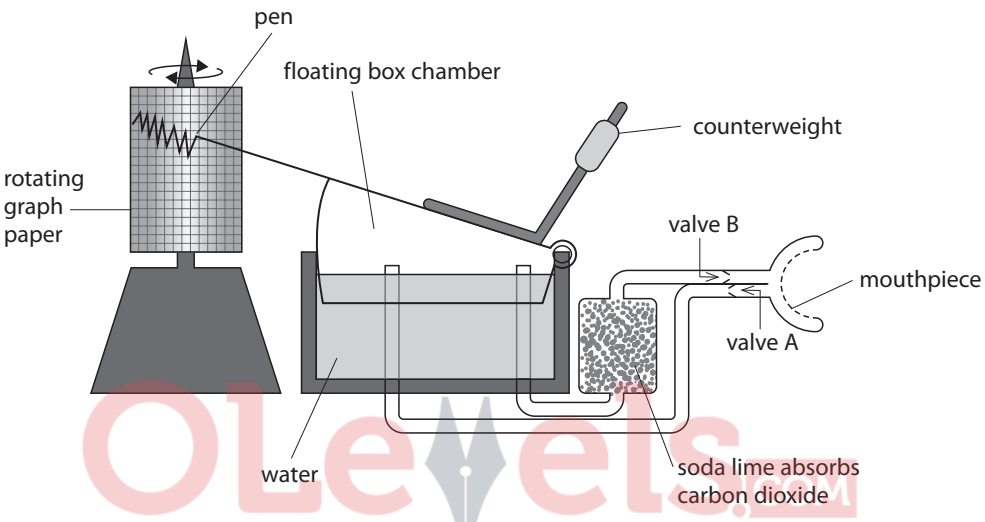
2. Question 6 · part a · 1 mark

January 2020 Paper 1

Further practice · Attempt part a only.

(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

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2. Question 6 · part a · 1 mark · Working and answer

Working and answer

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3. Question 2 · part a.iii · 1 mark

January 2022 Paper 2

Further practice · Attempt part a.iii only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 dm^3 of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

OLevels.COM — SCHOOL —

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

(3)

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3. Question 2 · part a.iii · 1 mark

January 2022 Paper 2

Further practice · Attempt part a.iii only.

(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

(2)

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

(1)

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

(1)

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

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3. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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4. Question 2 · part a.i · 2 marks

June 2023 Paper 1

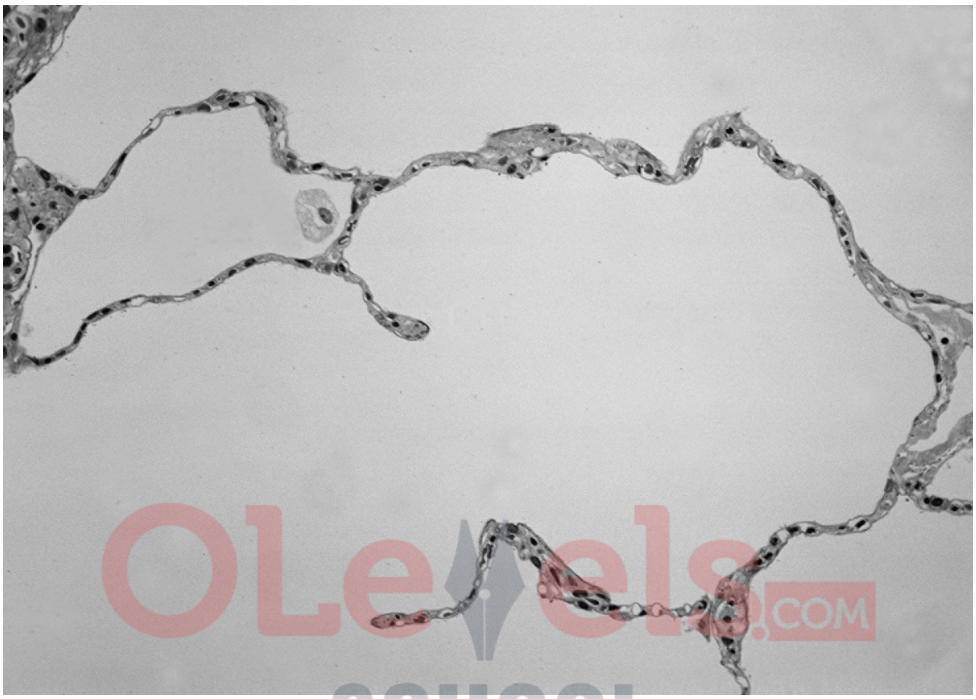
Further practice · Attempt part a.i only.

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2 The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

(2)

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

(2)

1 .....

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2 .....

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4. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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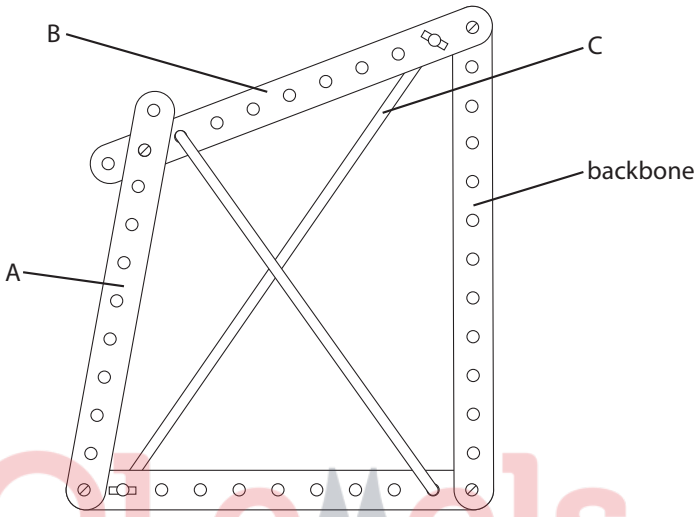
5. Question 5 · part a.i · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.i only.

5 (a) The diagram shows a model that can be used to demonstrate the process of breathing.

The part that represents the backbone has been labelled.



(i) State which parts of the breathing system are represented by A, B and C.

(3)

A .....

B .....

C .....

(ii) State three other structures, not shown in the diagram, that are needed for breathing in the body.

(3)

1 .....

2 .....

3 .....

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5. Question 5 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

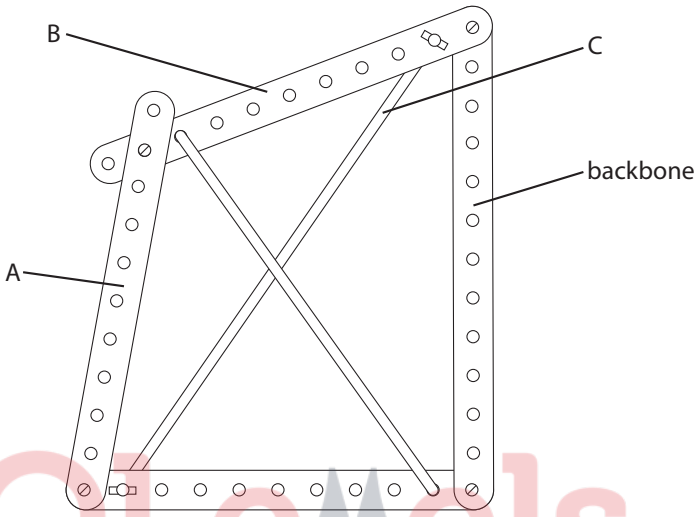
6. Question 5 · part a.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

5 (a) The diagram shows a model that can be used to demonstrate the process of breathing.

The part that represents the backbone has been labelled.



(i) State which parts of the breathing system are represented by A, B and C.

(3)

A .....

B .....

C .....

(ii) State three other structures, not shown in the diagram, that are needed for breathing in the body.

(3)

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2 .....

3 .....

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6. Question 5 · part a.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

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(iii) Describe how the structures present in the thorax are used to allow a person to inhale.

(6)

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(b) Give the balanced chemical symbol equation for the process of aerobic respiration.

(3)

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(Total for Question 5 = 15 marks)

6. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

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7. Question 4 · part b.i · 1 mark

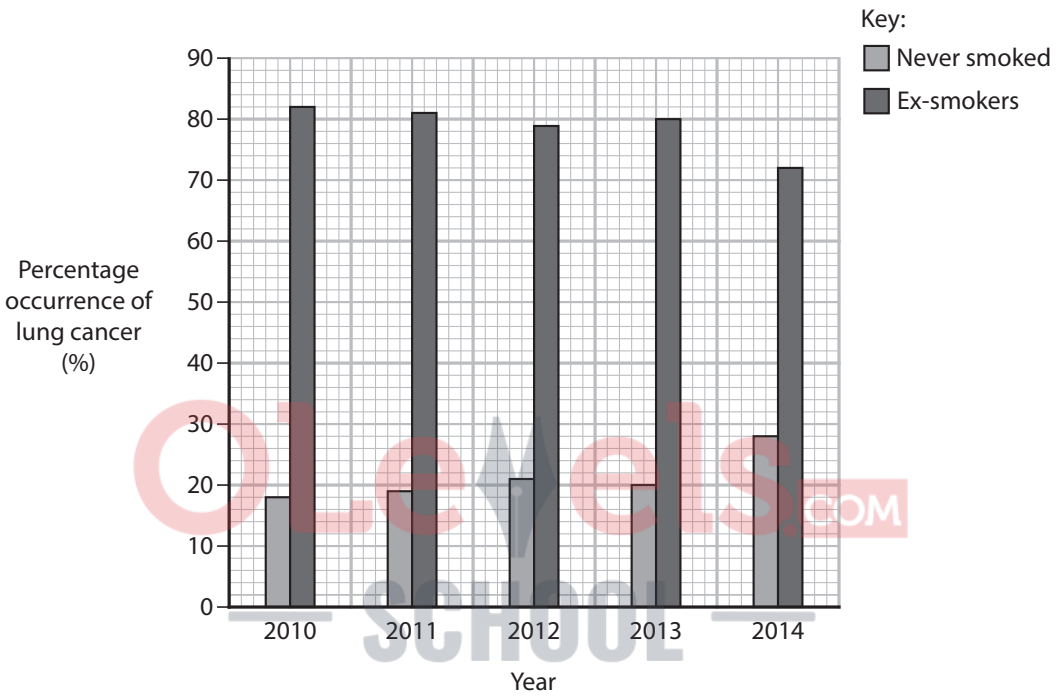
June 2025 Paper 2R

Further practice · Attempt part b.i only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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7. Question 4 · part b.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.i only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

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(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

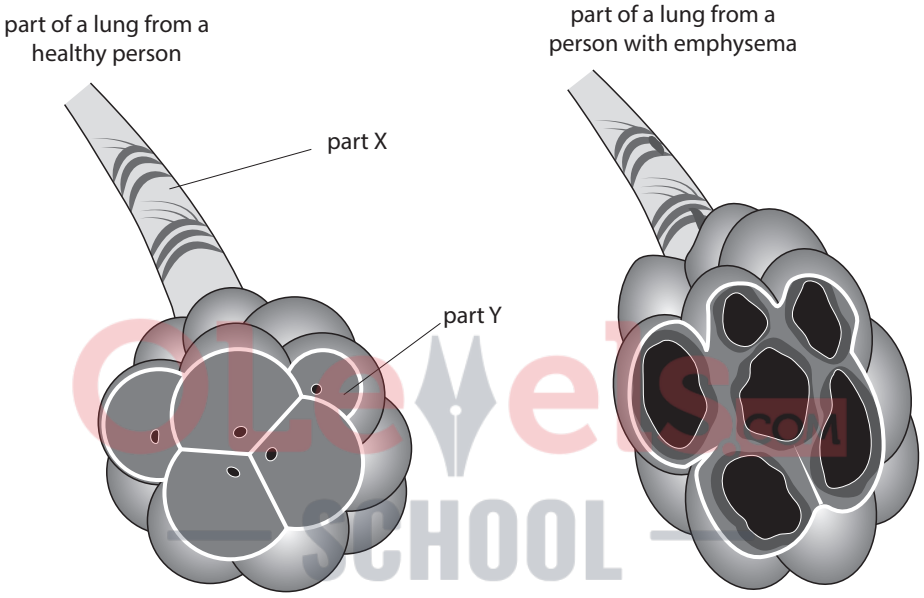
7. Question 4 · part b.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.i only.

(b) Emphysema is a lung condition that can be caused by cigarette smoking.

The diagram shows part of a lung from a healthy person and part of a lung from a person with emphysema.



(Source: © ttsz / Getty Images)

(i) Name part X.

(1)

.....

(ii) Name part Y.

(1)

.....

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7. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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8. Question 4 · part b.ii · 1 mark

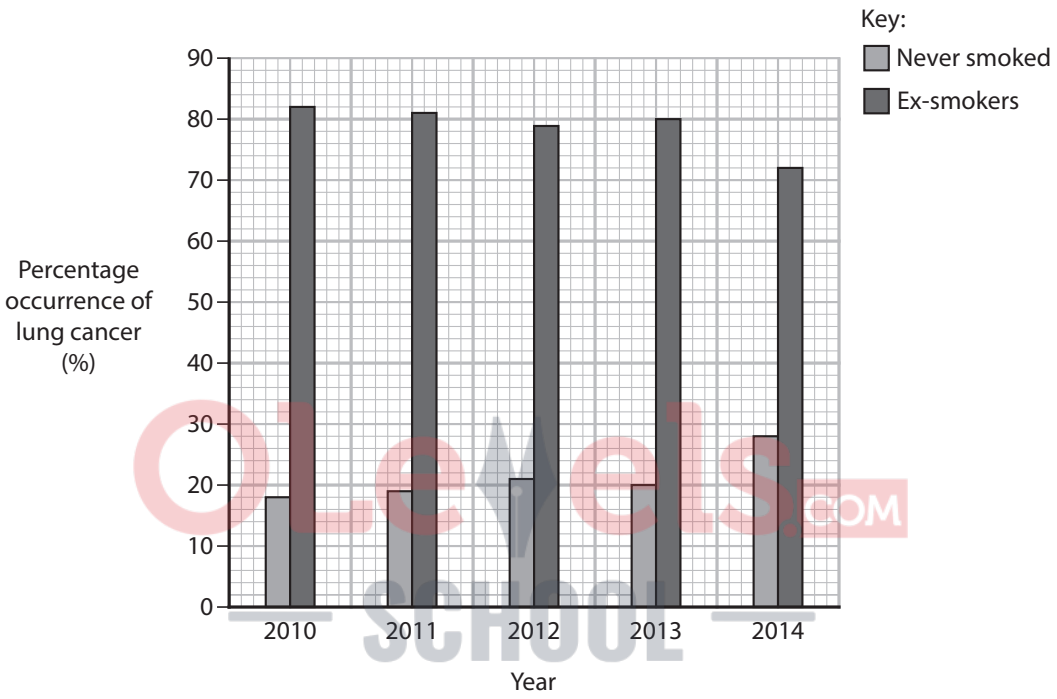
June 2025 Paper 2R

Further practice · Attempt part b.ii only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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8. Question 4 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

OLevels.COM SCHOOL

(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

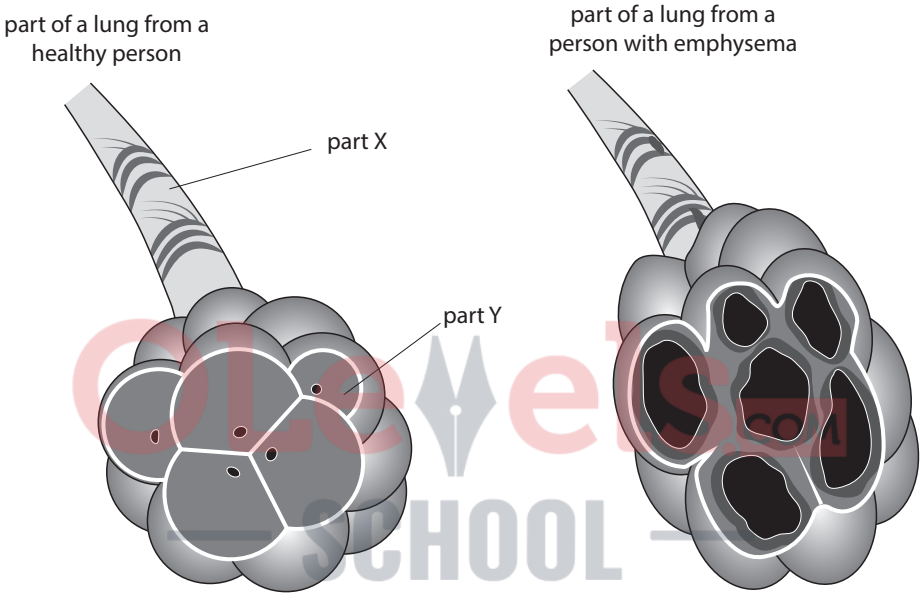
8. Question 4 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(b) Emphysema is a lung condition that can be caused by cigarette smoking.

The diagram shows part of a lung from a healthy person and part of a lung from a person with emphysema.



(Source: © ttsz / Getty Images)

(i) Name part X.

(1)

(ii) Name part Y.

(1)

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8. Question 4 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Explain the effect that emphysema has on the health of a person.

(4)

(Total for Question 4 = 13 marks)

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Working and answer

Practice workbook · Explain how diaphragm and intercostal muscles drive breathing

Muscle contraction changing thorax volume and pressure to move air, and what stops it.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VENTILATION-MUSCLE-ACTION

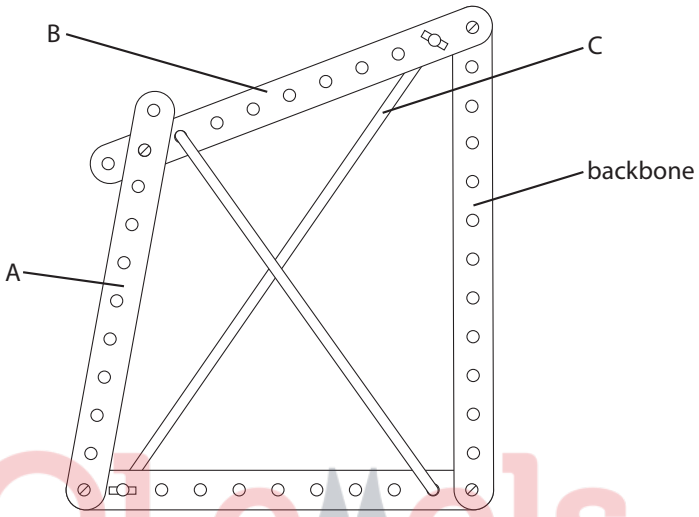
1. Question 5 · part a.iii · 6 marks

June 2025 Paper 1

Worked-example retry · Attempt part a.iii only.

5 (a) The diagram shows a model that can be used to demonstrate the process of breathing.

The part that represents the backbone has been labelled.



(i) State which parts of the breathing system are represented by A, B and C.

(3)

A .....

B .....

C .....

(ii) State three other structures, not shown in the diagram, that are needed for breathing in the body.

(3)

1 .....

2 .....

3 .....

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1. Question 5 · part a.iii · 6 marks

June 2025 Paper 1

Worked-example retry · Attempt part a.iii only.

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(iii) Describe how the structures present in the thorax are used to allow a person to inhale.

(6)

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(b) Give the balanced chemical symbol equation for the process of aerobic respiration.

(3)

..... + ... → ..... + ... + ...

(Total for Question 5 = 15 marks)

1. Question 5 · part a.iii · 6 marks · Working and answer

Working and answer

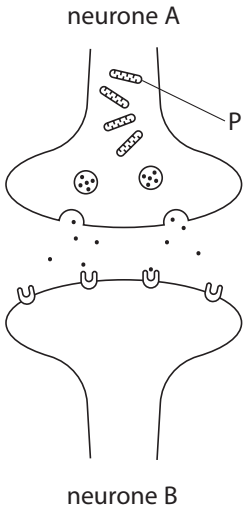
OLevels.com SCHOOL

2. Question 8 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

8 The diagram shows the gap between two neurones, A and B.



(a) (i) Draw an arrow on the diagram to show the direction of travel of an impulse.

(1)

(ii) Explain the function of structure P in the transmission of the impulse across the gap.

(3)

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2. Question 8 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

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(b) Curare is a poison used by South American Indians when hunting. Curare blocks receptors on the post synaptic membranes of neurones supplying skeletal muscles and prevents breathing.

(i) Explain how curare can stop the process of breathing.

(2)

(ii) Acetylcholinesterase is an enzyme found in the gap between two neurones.

Suggest why acetylcholinesterase inhibitors can be used to reverse the effects of curare poisoning.

(2)

(Total for Question 8 = 8 marks)

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2. Question 8 · part b.i · 2 marks · Working and answer

Working and answer

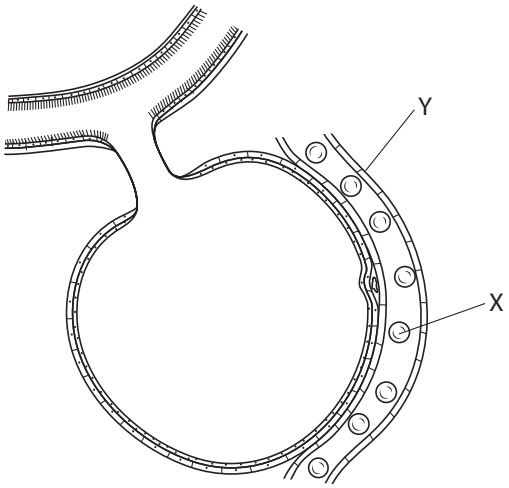
OLevels.com SCHOOL

3. Question 3 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

3 The diagram shows an alveolus and its blood supply.



(a) (i) What is the name of blood component X?

(1)

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere.

(3)

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2 .....

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3 .....

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3. Question 3 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

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(iii) Give three features of alveoli that allow efficient gas exchange.

(3)

1 .....

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2 .....

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(b) Structure Y is a capillary.

Give two features in the diagram which show that structure Y is a capillary.

(2)

1 .....

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2 .....

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(c) Some people have a condition known as emphysema. One symptom of emphysema is the breakdown of elastic fibres in the lung tissue.

Suggest the effects that loss of elastic fibres have on a person's ability to breathe.

(2)

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(Total for Question 3 = 11 marks)

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3. Question 3 · part c · 2 marks · Working and answer

Working and answer

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4. Question 7 · part a · 4 marks

June 2024 Paper 1

Further practice · Attempt part a only.

7 (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

(4)

(b) Describe the role of chemoreceptors in the carotid arteries.

(2)

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4. Question 7 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how smoking damages health, including in pregnancy

Tar, carbon monoxide and other smoke effects on lungs, circulation and the fetus, and reasons for lung cancer patterns.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SMOKING-HEALTH-DAMAGE

1. Question 2 · part d · 4 marks

January 2022 Paper 2

Worked-example retry · Attempt part d only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 dm^3 of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

(3)

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1. Question 2 · part d · 4 marks

January 2022 Paper 2

Worked-example retry · Attempt part d only.

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(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

(2)

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

(1)

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

OLevels.COM SCHOOL

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

(1)

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

1. Question 2 · part d · 4 marks

January 2022 Paper 2

Worked-example retry · Attempt part d only.

(c) Calculate the average volume of oxygen absorbed by the lungs during one minute of breathing, at rest. (lines 13 to 15)

(3)

volume of oxygen absorbed = ... dm^3

(d) Explain why pregnant women are advised not to smoke cigarettes.

(4)

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(Total for Question 2 = 14 marks)

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1. Question 2 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part c · 5 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

3 A woman who smokes when she is pregnant risks harming her developing fetus.

The table lists the birth masses of 310 children born to non-smoking mothers or to mothers who smoke.

	Birth masses of children in kg					
	Less than 2.4	**2.4 – 3.1**	**3.2 – 3.5**	**3.6 – 3.9**	**More than 3.9**	**Total number of births in sample**
Number of children born to non-smoking mothers	4	42	70	60	34	210
Number of children born to mothers who smoke	3	24	44	23	6	100

(a) (i) What is the most common birth mass in kg?

(1)

- ☒ **A** 2.4 – 3.1
- ☒ **B** 3.2 – 3.5
- ☒ **C** 3.6 – 3.9
- ☒ **D** More than 3.9

(ii) Calculate the percentage of children with a birth mass of 3.6 kg or more in the sample of non-smoking mothers.

(2)

percentage = ...

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2. Question 3 · part c · 5 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

(b) Use data from the table to deduce what can be concluded about the effect of smoking on birth mass.

(2)

(c) Suggest why smoking affects birth mass.

(5)

(Total for Question 3 = 10 marks)

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2. Question 3 · part c · 5 marks · Working and answer

Working and answer

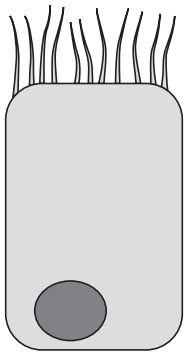
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3. Question 2 · part b.i · 1 mark

November 2020 Paper 1R

Further practice · Attempt part b.i only.

2 The diagram shows a cell from the human breathing system.



(a) (i) Add these labels to the diagram.

- cilia
- cell membrane
- nucleus

(3)

(ii) Each of the cell structures has a particular function.

The boxes give a list of structures and functions.

Draw one straight line from each structure to its function.

(3)

Structure	Function
cilia	holds the genetic code
cell membrane	site where most chemical reactions occur
nucleus	controls the entry and exit of substances
	makes proteins
	moves mucus through the trachea

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3. Question 2 · part b.i · 1 mark

November 2020 Paper 1R

Further practice · Attempt part b.i only.

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(b) (i) Cigarette smoking affects the function of the cell shown in the diagram.

Name a substance in cigarette smoke that causes damage to this cell.

(1)

.....

(ii) Describe how cigarette smoking affects the cell shown in the diagram.

(2)

.....

.....

.....

.....

(Total for Question 2 = 9 marks)

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Working and answer

4. Question 3 · part a · 2 marks

June 2024 Paper 2

Further practice · Attempt part a only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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4. Question 3 · part a · 2 marks

June 2024 Paper 2

Further practice · Attempt part a only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

4. Question 3 · part a · 2 marks · Working and answer

Working and answer

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5. Question 4 · part a.iii · 2 marks

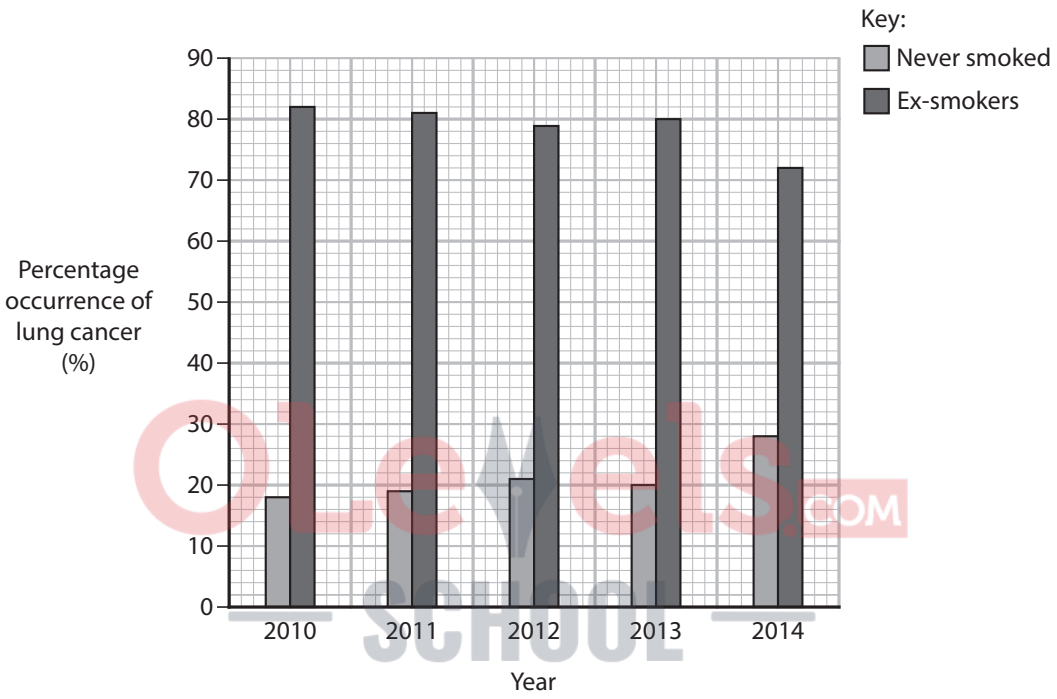
June 2025 Paper 2R

Further practice · Attempt part a.iii only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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5. Question 4 · part a.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.iii only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

.....

.....

.....

.....

(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

.....

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5. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

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6. Question 4 · part a.iv · 1 mark

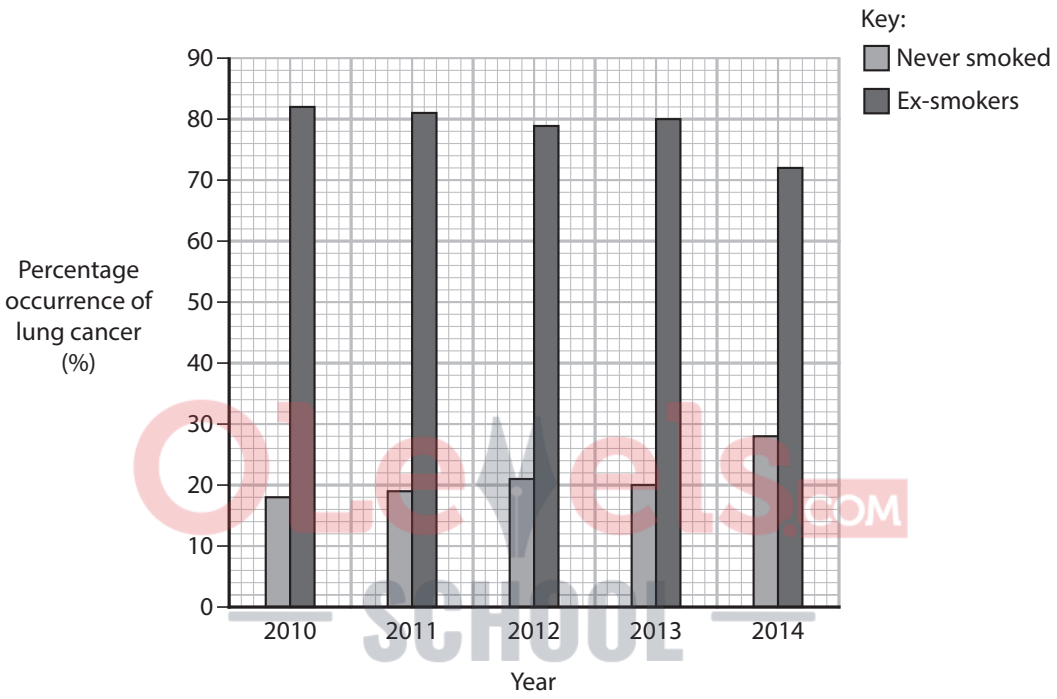
June 2025 Paper 2R

Further practice · Attempt part a.iv only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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6. Question 4 · part a.iv · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.iv only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

.....

.....

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.....

(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

.....

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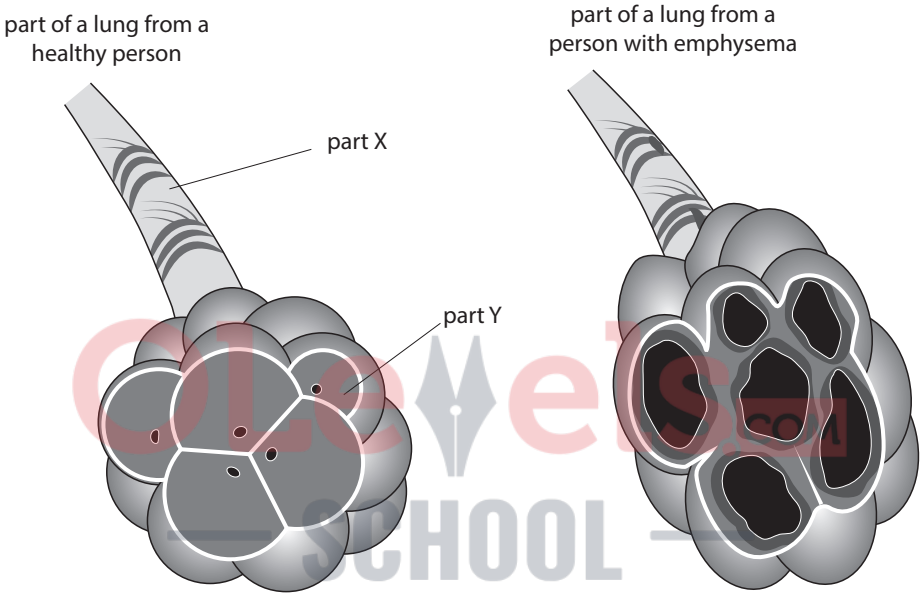
6. Question 4 · part a.iv · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.iv only.

(b) Emphysema is a lung condition that can be caused by cigarette smoking.

The diagram shows part of a lung from a healthy person and part of a lung from a person with emphysema.



(Source: © ttsz / Getty Images)

(i) Name part X.

(1)

(ii) Name part Y.

(1)

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P 7 4 6 3 2 A 0 1 2 2 4

6. Question 4 · part a.iv · 1 mark · Working and answer

Working and answer

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Practice workbook · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Raised heart and breathing rate, blood redirection, adrenaline, nerve control, recovery and oxygen debt, and chemoreceptors and the medulla responding to carbon dioxide.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXERCISE-AND-BREATHING-CONTROL

1. Question 5 · part b.i · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.i only.

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5 (a) A red blood cell travelling from the liver to the kidney passes through six different arteries and veins.

Complete the flow chart by naming these blood vessels in order. The first one has been done for you.

(5)

LIVER

↓

hepatic vein

↓

.....

↓

.....

↓

.....

↓

.....

↓

.....

↓

KIDNEY

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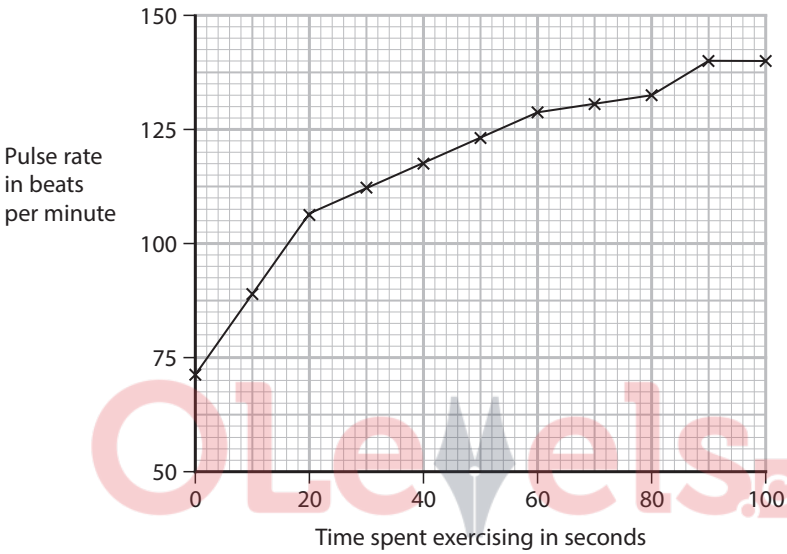
1. Question 5 · part b.i · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.i only.

(b) A student investigates the effect of time spent exercising on pulse rate.

The graph shows the student's results.



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1. Question 5 · part b.i · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.i only.

(i) Explain the change in pulse rate recorded by the student.

(4)

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1. Question 5 · part b.i · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.i only.

(ii) Describe how the student could carry out the investigation.

(6)

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(Total for Question 5 = 15 marks)

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1. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

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2. Question 8 · part c · 4 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

8 The table shows the rate of blood flow through different organs when a person is at rest. The table also shows the masses of these organs.

Organ	Mass in kg	Rate of blood flow in cm^3 per minute	
		through the whole organ	per 100g of the organ
kidneys	0.3	1200	
skeletal muscles	25.0		3
liver	1.5	1500	100

(a) Calculate the missing values and put them in the table.

(4)

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(b) Explain why the liver has a high rate of blood flow.

(4)

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2. Question 8 · part c · 4 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

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(c) Explain any changes that would occur in the blood flow through these organs during vigorous exercise.

(4)

(Total for Question 8 = 12 marks)

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Working and answer

3. Question 9 · part a.ii · 4 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

9 In sports such as archery and shooting, beta-blockers are used illegally by some competitors to help reduce the effects of stress and anxiety, which can affect performance.

(a) (i) Suggest why the use of beta-blockers in competitive sport is illegal.

(1)

(ii) Stress and anxiety can cause an increase in the release of adrenaline.

Explain the effects of adrenaline on the body.

(4)

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3. Question 9 · part a.ii · 4 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

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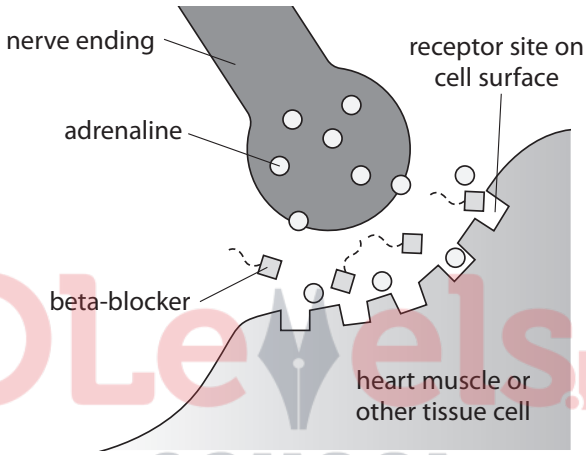
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(b) Adrenaline acts as a hormone and as a neurotransmitter.

Beta-blockers act at the junction between nerve endings and body tissues where adrenaline is found.

The diagram shows the action of beta-blockers at a synapse.



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(i) Explain how beta-blockers reduce symptoms of stress and anxiety.

(3)

(ii) Suggest how the action of beta-blockers would be a disadvantage in active sports such as athletics.

(2)

(Total for Question 9 = 10 marks)

3. Question 9 · part a.ii · 4 marks · Working and answer

Working and answer

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4. Question 2 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

2 A student records her pulse rate in beats per minute (bpm) after different activities.

These are the student's results.

- at rest 78 bpm
- walking 95 bpm
- jogging 138 bpm
- running 146 bpm

(a) Draw a table to show the student's results.

(3)

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4. Question 2 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

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(b) Describe a method that the student could use to obtain these results.

(4)

(c) State one factor that the student should control during the investigation.

(1)

(d) Name a body process, other than pulse rate, that increases during exercise.

(1)

(Total for Question 2 = 9 marks)

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4. Question 2 · part d · 1 mark · Working and answer

Working and answer

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5. Question 5 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
atropine injected → 9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

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5. Question 5 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

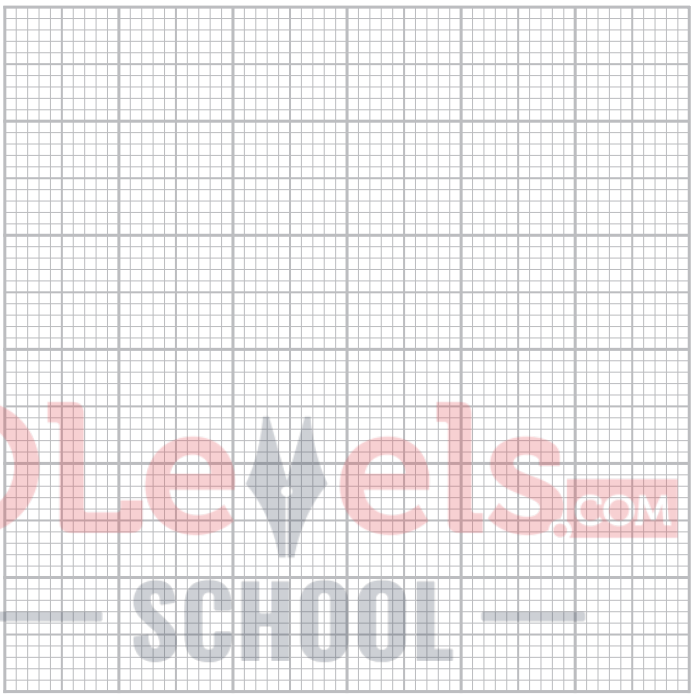
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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



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(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

.....

.....

.....

.....

(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

5. Question 5 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

(ii) Explain why this percentage change is not achieved as soon as atropine is injected.

(2)

.....

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.....

(c) Suggest the function of the vagus nerve.

(2)

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(Total for Question 5 = 14 marks)

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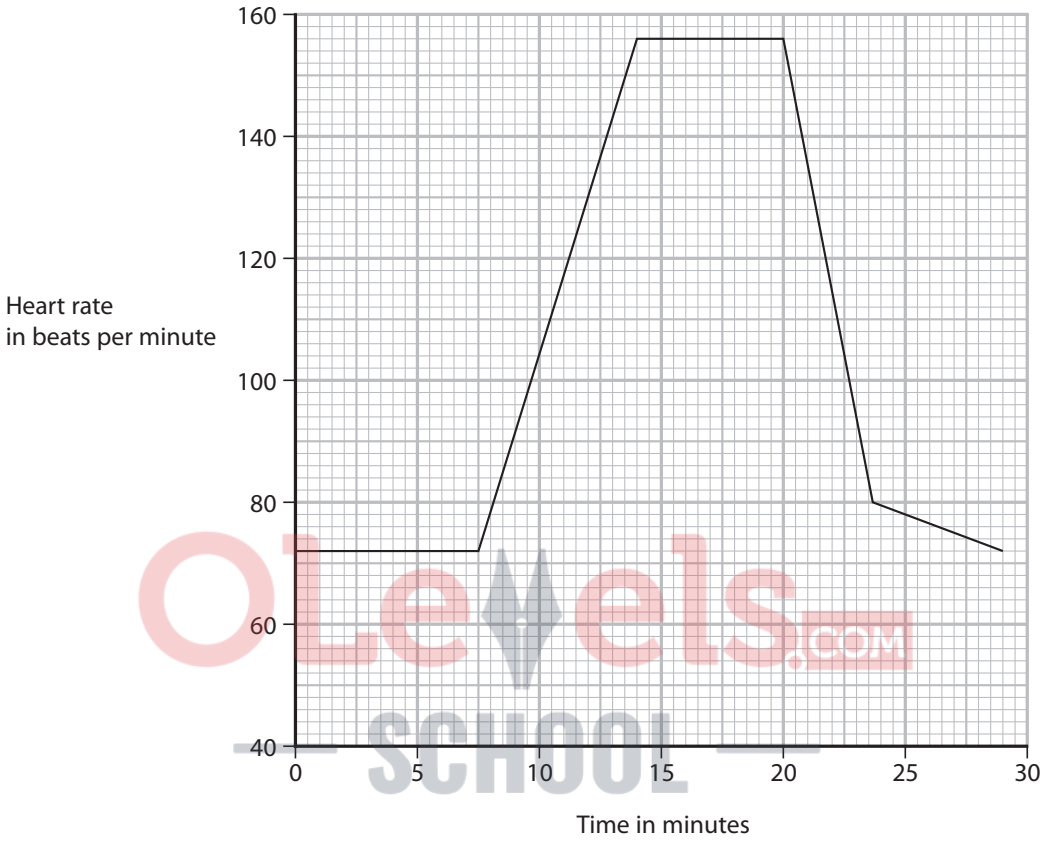
Working and answer

6. Question 2 · part a.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.ii only.

2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

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6. Question 2 · part a.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.ii only.

(iii) Complete the passage about what happens in the body during exercise using the correct words.

(4)

During exercise ..... gas moves from the lungs into the blood by a process called ..... . This gas is needed by cells for ..... respiration which releases energy. Also during exercise, the amount of urine produced ..... as water is lost in sweat.

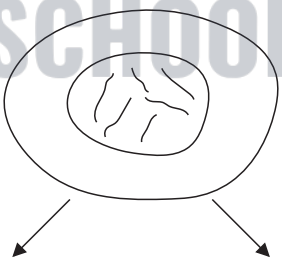
(b) (i) During exercise, body cells get damaged.

Body cells are replaced by a process called mitosis.

The diagram represents one body cell but shows only 6 of the 46 chromosomes.

Complete the diagram to show the new body cells produced from this cell by mitosis.

(2)



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6. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

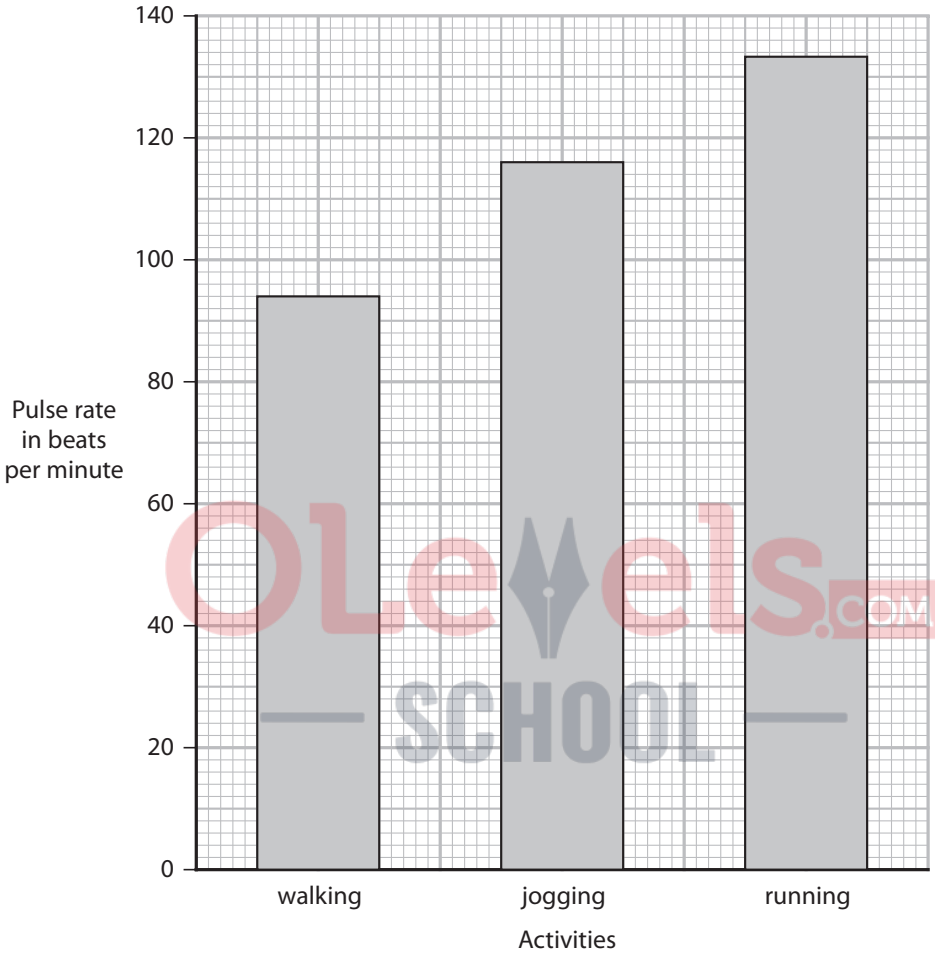
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7. Question 4 · part b · 4 marks

June 2023 Paper 2R

Further practice · Attempt part b only.

4 (a) The bar chart shows the pulse rate of a student at the end of three different activities.



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7. Question 4 · part b · 4 marks

June 2023 Paper 2R

Further practice · Attempt part b only.

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(i) Draw a table that the student could use to record these results.

(3)

(ii) Complete the table by adding the student's results.

(2)

(iii) Describe a method that the student could use to obtain the results shown in the bar chart.

(4)

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7. Question 4 · part b · 4 marks

June 2023 Paper 2R

Further practice · Attempt part b only.

(b) The pulse rates at the end of walking and running are different.

Explain this difference.

(4)

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(Total for Question 4 = 13 marks)

7. Question 4 · part b · 4 marks · Working and answer

Working and answer

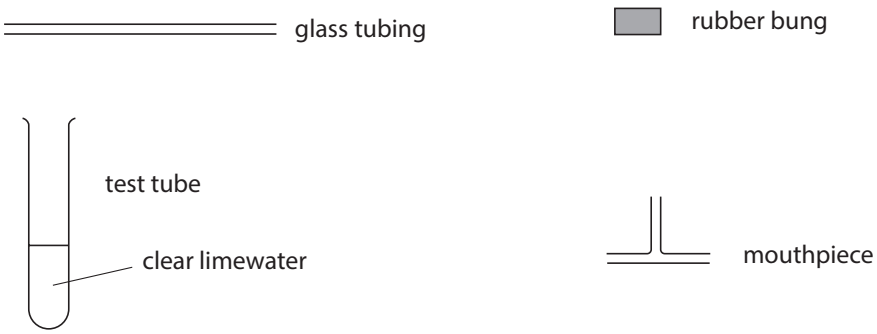
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8. Question 4 · part b.ii · 4 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

4 (a) The diagram shows some scientific equipment.



(i) Draw a diagram to show how this equipment is set up to investigate the difference between the carbon dioxide concentration of inspired air and of expired air.

(3)

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8. Question 4 · part b.ii · 4 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

(ii) State one safety precaution in this investigation.

(1)

(b) A student investigates their breathing rate at rest, after gentle exercise and again after vigorous exercise.

The student repeats each exercise three times.

The student's mean results are shown in the table.

Mean breathing rate in breaths per minute		
at rest	**after gentle exercise**	**after vigorous exercise**
14	19	50

(i) Give one control variable in this investigation.

(1)

(ii) Explain how the breathing rate after vigorous exercise benefits working muscles.

(4)

(Total for Question 4 = 9 marks)

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8. Question 4 · part b.ii · 4 marks · Working and answer

Working and answer

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9. Question 2 · part b · 1 mark

January 2022 Paper 2

Further practice · Attempt part b only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 dm^3 of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

(3)

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9. Question 2 · part b · 1 mark

January 2022 Paper 2

Further practice · Attempt part b only.

(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

(2)

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

(1)

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

(1)

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

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9. Question 2 · part b · 1 mark

January 2022 Paper 2

Further practice · Attempt part b only.

(c) Calculate the average volume of oxygen absorbed by the lungs during one minute of breathing, at rest. (lines 13 to 15)

(3)

volume of oxygen absorbed = ... dm^3

(d) Explain why pregnant women are advised not to smoke cigarettes.

(4)

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(Total for Question 2 = 14 marks)

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9. Question 2 · part b · 1 mark · Working and answer

Working and answer

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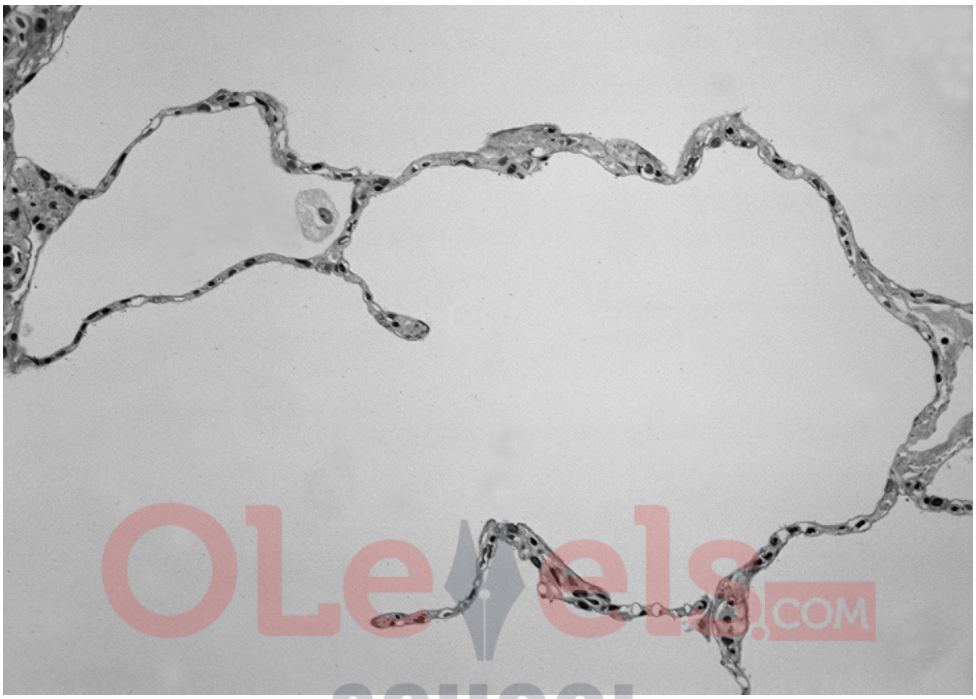
10. Question 2 · part b.iv · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iv only.

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2 The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

(2)

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

(2)

1 .....

.....

2 .....

.....

10. Question 2 · part b.iv · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iv only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

(1)

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

Percentage (%) carbon dioxide in air	0.02	1.60	3.00	6.00
Tidal volume in cm^3	500	700	1200	2100
Rate of breathing in breaths per minute	14	15	16	28

Calculate the volume of air in dm^3 breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 dm^3 = 1000 cm^3]

(2)

volume of air = ... dm^3

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

(3)

volume of oxygen = ... cm^3

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10. Question 2 · part b.iv · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iv only.

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(iv) Explain the effect of increasing the percentage of carbon dioxide in the air a person breathes in on the tidal volume and rate of breathing.

(4)

(Total for Question 2 = 14 marks)

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Working and answer

11. Question 7 · part b · 2 marks

June 2024 Paper 1

Further practice · Attempt part b only.

7 (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

(4)

(b) Describe the role of chemoreceptors in the carotid arteries.

(2)

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11. Question 7 · part b · 2 marks

June 2024 Paper 1

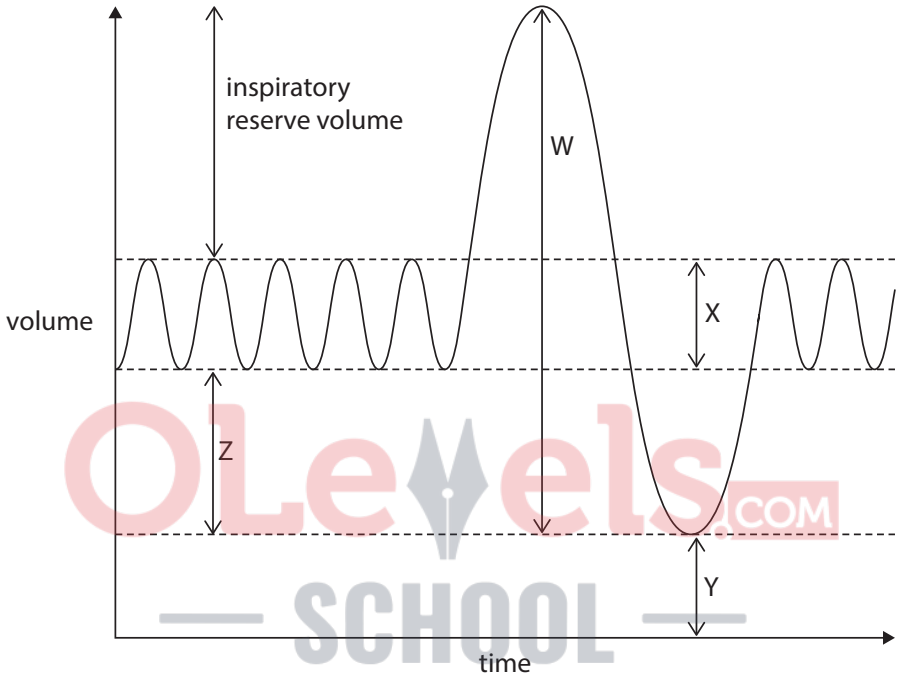
Further practice · Attempt part b only.

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(c) The graph shows different lung volumes.



(Source: https://www.inds.co.uk/education/how2/measure-lung-volume/)

(i) Give the name of the lung volume represented by letter X.

(1)

.....

(ii) Give the name of the lung volume represented by letter Y.

(1)

.....

(iii) Give the name of the lung volume represented by letter W.

(1)

.....

(Total for Question 7 = 9 marks)

11. Question 7 · part b · 2 marks · Working and answer

Working and answer

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Practice workbook · Explain how alveoli are adapted for gas exchange and effects of damage

Alveolar features for diffusion, differences between alveolar and outside air, and how emphysema reduces exchange.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ALVEOLI-GAS-EXCHANGE

1. Question 4 · part b.iii · 4 marks

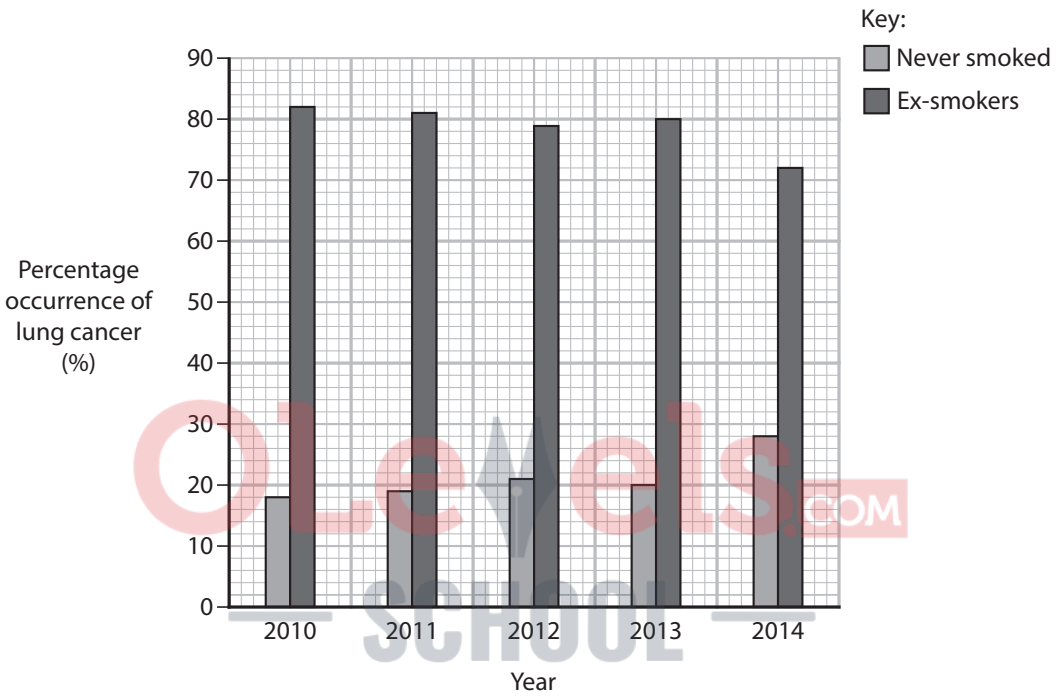
June 2025 Paper 2R

Worked-example retry · Attempt part b.iii only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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1. Question 4 · part b.iii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.iii only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

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(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

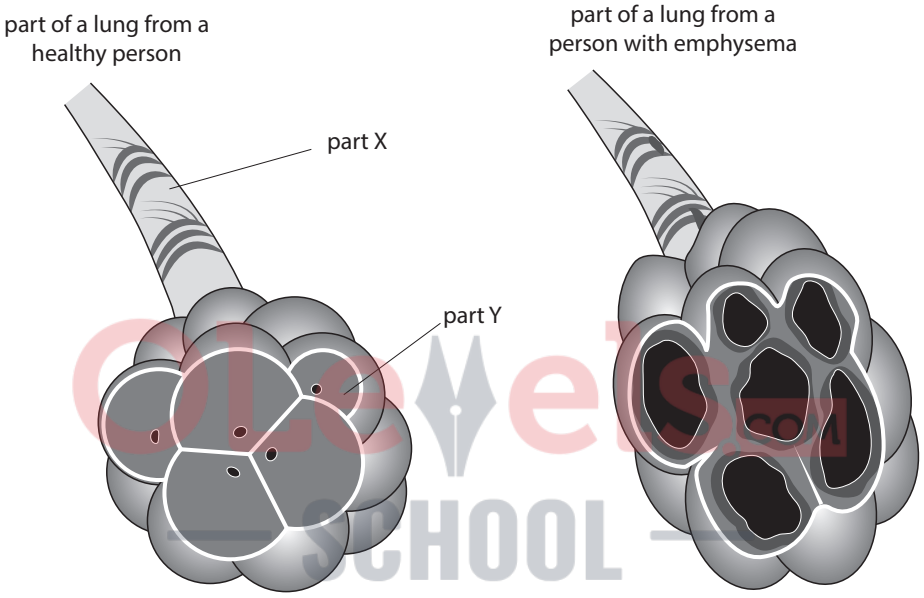
1. Question 4 · part b.iii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.iii only.

(b) Emphysema is a lung condition that can be caused by cigarette smoking.

The diagram shows part of a lung from a healthy person and part of a lung from a person with emphysema.



(Source: © ttsz / Getty Images)

(i) Name part X.

(1)

(ii) Name part Y.

(1)

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1. Question 4 · part b.iii · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.iii only.

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(iii) Explain the effect that emphysema has on the health of a person.

(4)

(Total for Question 4 = 13 marks)

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Working and answer

2. Question 6 · part c.ii · 3 marks

June 2019 Paper 1

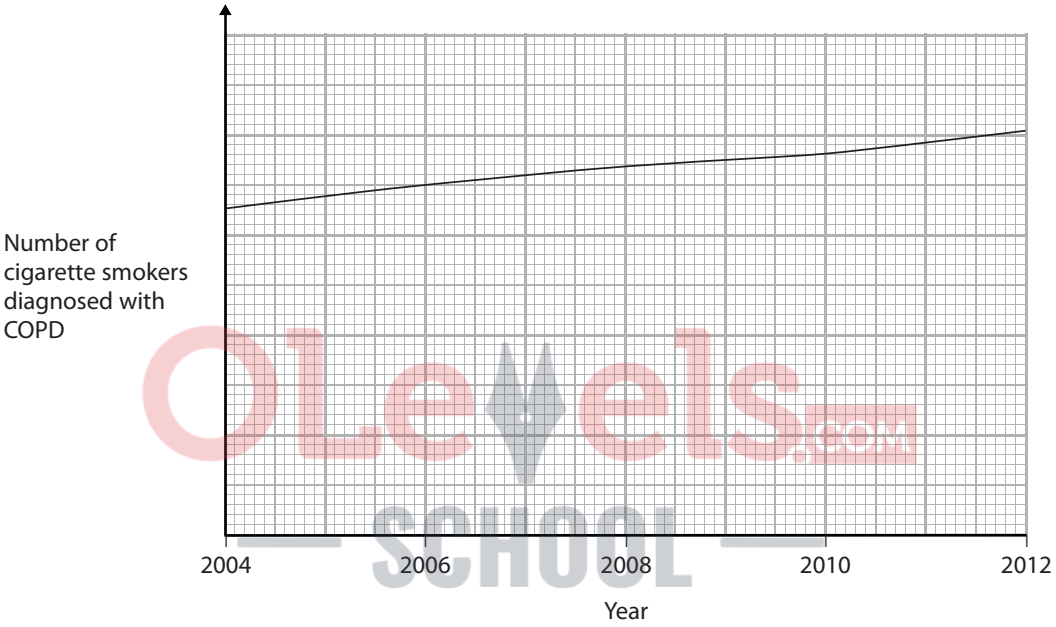
Further practice · Attempt part c.ii only.

6 (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.



(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

(1)

(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

(2)

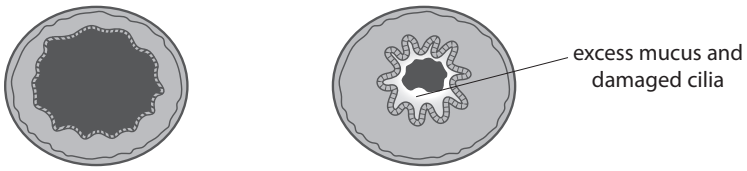
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2. Question 6 · part c.ii · 3 marks

June 2019 Paper 1

Further practice · Attempt part c.ii only.

(b) The diagram shows how chronic bronchitis affects the airways in the breathing system.



Explain how excess mucus and damaged cilia affect the breathing system of a person with chronic bronchitis.

(2)

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2. Question 6 · part c.ii · 3 marks

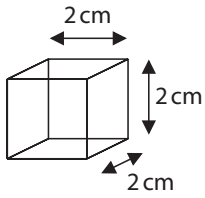
June 2019 Paper 1

Further practice · Attempt part c.ii only.

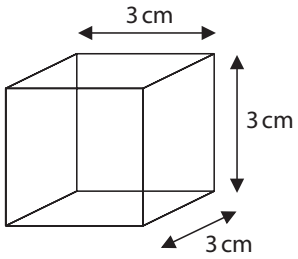
(c) The diagram shows models of two alveoli.

One model represents an alveolus from a person with healthy lungs.

The other model represents an alveolus from a person with emphysema.



Healthy



Emphysema

The table shows the surface area to volume ratio for a healthy alveolus.

	Surface area in cm^2	Volume in cm^3	Surface area to volume ratio
Healthy	24	8	3 : 1
Emphysema			

(i) Complete the table by giving the missing information.

(3)

(ii) Explain how the surface area to volume ratio of alveoli in the lungs of a person with emphysema will affect the normal function of body cells.

(3)

(Total for Question 6 = 11 marks)

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2. Question 6 · part c.ii · 3 marks · Working and answer

Working and answer

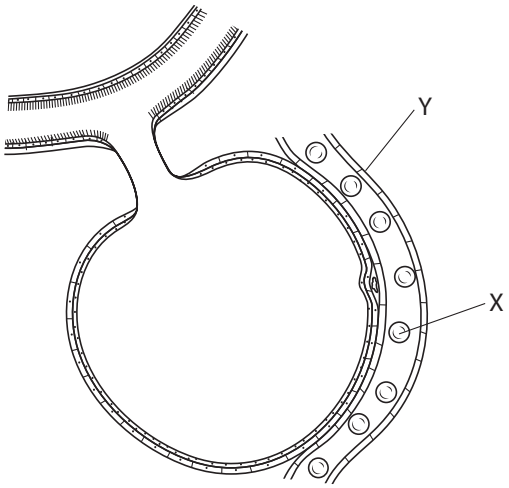
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3. Question 3 · part a.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

3 The diagram shows an alveolus and its blood supply.



(a) (i) What is the name of blood component X? **(1)**

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere. **(3)**

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3. Question 3 · part a.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

(iii) Give three features of alveoli that allow efficient gas exchange.

(3)

1 .....

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2 .....

.....

3 .....

.....

(b) Structure Y is a capillary.

Give two features in the diagram which show that structure Y is a capillary.

(2)

1 .....

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2 .....

.....

(c) Some people have a condition known as emphysema. One symptom of emphysema is the breakdown of elastic fibres in the lung tissue.

Suggest the effects that loss of elastic fibres have on a person's ability to breathe.

(2)

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(Total for Question 3 = 11 marks)

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3. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

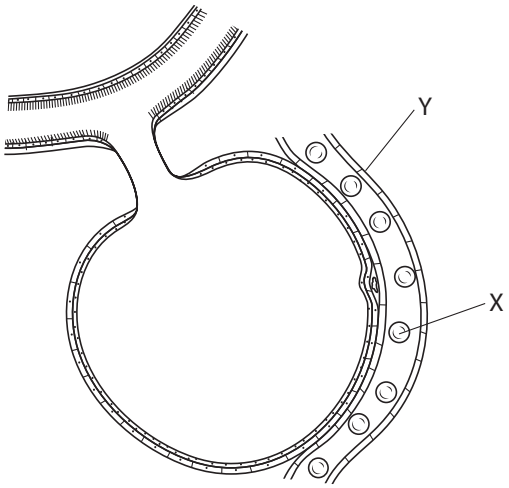
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4. Question 3 · part a.iii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.iii only.

3 The diagram shows an alveolus and its blood supply.



(a) (i) What is the name of blood component X? **(1)**

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere. **(3)**

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4. Question 3 · part a.iii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.iii only.

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(iii) Give three features of alveoli that allow efficient gas exchange.

(3)

1 .....

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2 .....

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3 .....

.....

(b) Structure Y is a capillary.

Give two features in the diagram which show that structure Y is a capillary.

(2)

1 .....

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2 .....

.....

(c) Some people have a condition known as emphysema. One symptom of emphysema is the breakdown of elastic fibres in the lung tissue.

Suggest the effects that loss of elastic fibres have on a person's ability to breathe.

(2)

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(Total for Question 3 = 11 marks)

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4. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

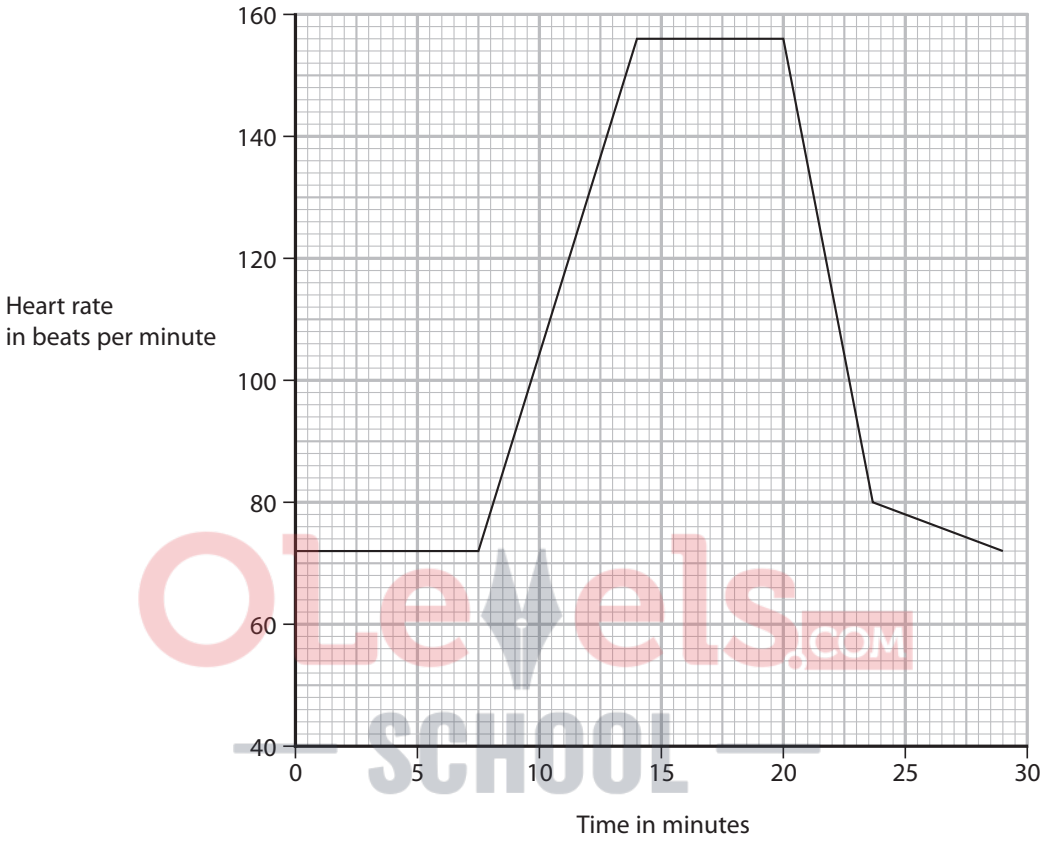
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5. Question 2 · part a.iii · 4 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

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5. Question 2 · part a.iii · 4 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

(iii) Complete the passage about what happens in the body during exercise using the correct words.

(4)

During exercise ..... gas moves from the lungs into the blood by a process called ..... . This gas is needed by cells for ..... respiration which releases energy. Also during exercise, the amount of urine produced ..... as water is lost in sweat.

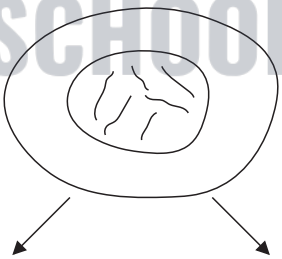
(b) (i) During exercise, body cells get damaged.

Body cells are replaced by a process called mitosis.

The diagram represents one body cell but shows only 6 of the 46 chromosomes.

Complete the diagram to show the new body cells produced from this cell by mitosis.

(2)



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5. Question 2 · part a.iii · 4 marks · Working and answer

Working and answer

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6. Question 2 · part a.ii · 2 marks

June 2023 Paper 1

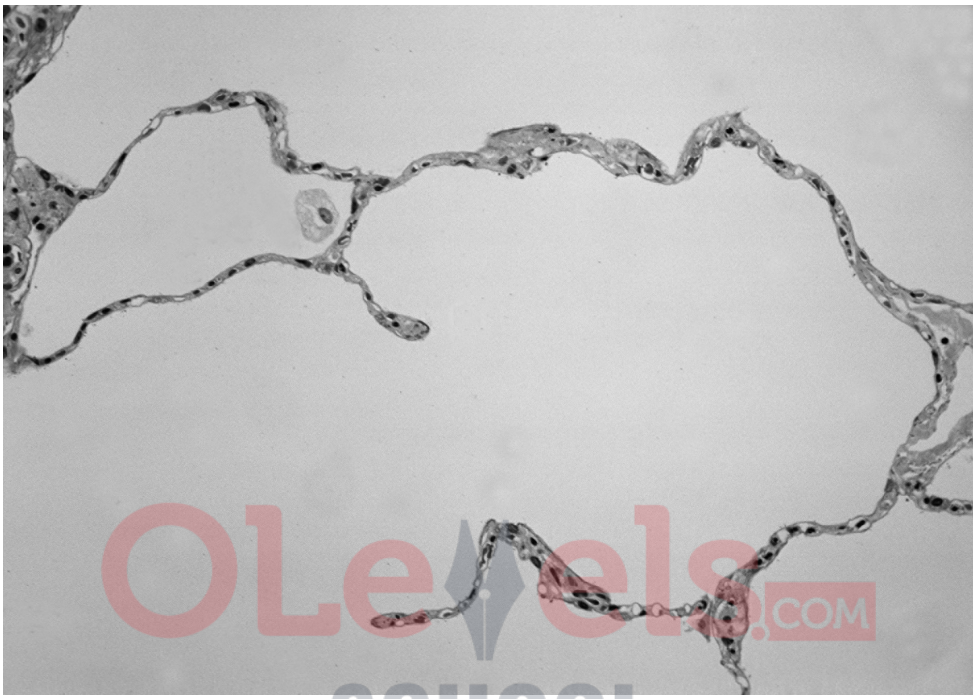
Further practice · Attempt part a.ii only.

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2 The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

(2)

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

(2)

1 .....

.....

2 .....

.....

6. Question 2 · part a.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.ii only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

(1)

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

Percentage (%) carbon dioxide in air	0.02	1.60	3.00	6.00
Tidal volume in cm^3	500	700	1200	2100
Rate of breathing in breaths per minute	14	15	16	28

Calculate the volume of air in dm^3 breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 dm^3 = 1000 cm^3]

(2)

volume of air = ... dm^3

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

(3)

volume of oxygen = ... cm^3

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6. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

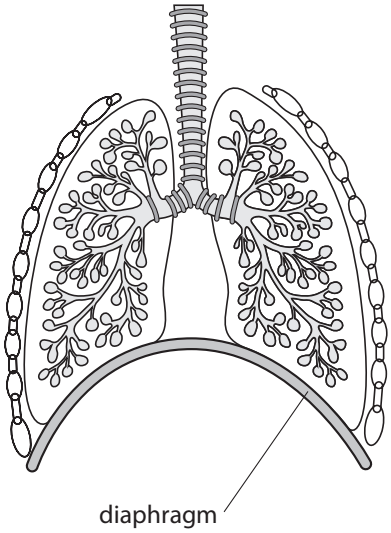
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7. Question 4 · part a.ii · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

4 (a) The diagram shows the human lungs.



(i) Add the following labels on the diagram

(3)

- trachea
- bronchus
- alveoli

(ii) Give four features of the lungs that adapt them for gas exchange.

(4)

1 .....

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2 .....

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3 .....

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4 .....

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7. Question 4 · part a.ii · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

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(b) A spirometer can be used to test lung function.

(i) Describe how a spirometer is used to obtain values for tidal volume.

(4)

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(ii) Asthma is a condition that causes the tubes in the lungs to narrow.

A person without asthma has a vital capacity of 4600 cm^3.

People with severe asthma have a 30% lower vital capacity compared with a healthy person.

Calculate the vital capacity of a person with severe asthma.

(2)

vital capacity of a person with severe asthma = ... cm^3

(Total for Question 4 = 13 marks)

7. Question 4 · part a.ii · 4 marks · Working and answer

Working and answer

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